

Deep Learning & AI in Sports

twelve¹²



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Dr. Pegah Rahimian

Dr. Pegah Rahimian is a football data scientist at Twelve Football and a post-doctoral researcher at Uppsala University. She holds a PhD in Football Data Science and has applied AI to derive in-depth analyses from football data, influencing various aspects of the sport.

October 2025

Mondays & Thursdays:
17:00 - 19:00 CET

twelve¹²

Professor David Sumpter

Professor David Sumpter is a professor of applied mathematics at Uppsala University and the author of "Soccermatics," a book that explores the application of mathematical modeling to football. He co-founded Twelve Football and has worked with major clubs and national federations worldwide.



Course Roadmap

The key is to move from **structure** → **interactions** → **valuation** → **communication**,



Session 1: Tracking → Team Shapes, defensive lines & Formation Clusters



Session 2: Playing-Style & Role Representations (clustering/embeddings)



Session 3: Graph Neural Networks for Pass Risk & Reward, Pitch Control



Session 4: Action Valuation I — Expected Threat (xT) & VAEP



Session 5: Action Valuation II — Pitch Control, EVP & Reinforcement Learning



Session 6: Interpretability & Explainable ML in Football



Session 7: Communicating With LLM Assistants (coach-ready insights)



Session 8: Project Studio & Demos

Module 5:

Valuation & Modern Scouting using Event + Tracking Data

Agenda

01

Pitch Control

02

EPV

03

Soccermap

04

Reinforcement Learning



From Events to Space and Sequences

- Traditional event-based models (xT, VAEP, OBV) only describe *what happened* with the ball.
- Modern models integrate *spatial control* and *future sequences*, answering:

“Who controlled the game state before an action, and how did it change after?”



Pitch Control Models

 **MIT SLOAN**
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Beyond Expected Goals

William Spearman
Hudl
william.spearman@hudl.com

 **MIT SLOAN**
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presented by **ESPN**

Wide Open Spaces: A statistical technique for measuring space creation in professional soccer

Javier Fernandez
F.C. Barcelona
javier.fernandezr@fcbarcelona.cat

Luke Bornn
Simon Fraser University, Sacramento Kings
lbornn@sfu.ca

Pitch Control: The Spearman Model (2017)

PC at a particular location is the probability that the player could control the ball, assuming it were at that location

Based on “Physics-based” interception model using tracking data.

Each player has:

$$TTI_i(x, y) = \tau_{\text{reaction}} + t_{\text{move}}(d_{i,(x,y)}, v_i, a_i)$$

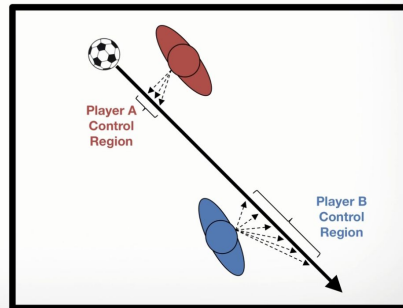
where:

- τ_{reaction} : reaction time delay (≈ 0.7 s)
- t_{move} : time to reach point (based on distance, velocity, and acceleration)

Control probability of team A over location (x, y):

$$P_A(x, y) = \frac{1}{1 + e^{-(TTI_B - TTI_A)/\sigma}}$$

- **Control Density**
 - 10% pass probability per intercept vector
- **Player A:**
 - Line 1: 10%
 - Line 2: (90%) x 10% = 9%
 - Line 3: (81%) x 10% = 8.1%
 - Total: 27.1% probability of controlling the pass
- **Player B:**
 - Line 1: (72.9%) x 10% = 7.3%
 - Line 2: (65.6%) x 10% = 6.6%
 - Line 3: (59.0%) x 10% = 5.9%
 - Line 4: (53.1%) x 10% = 5.3%
 - Line 5: (47.8%) x 10% = 4.8%
 - Line 6: (43.0%) x 10% = 4.3%
 - Total: 34.2% probability of controlling the pass



“Who gets there first?”

soft logistic “race” between attackers and defenders

Pitch Control: The Spearman Model (2017)

In football perspective, PC Measures “who is likely to reach the ball first” at any pitch location.

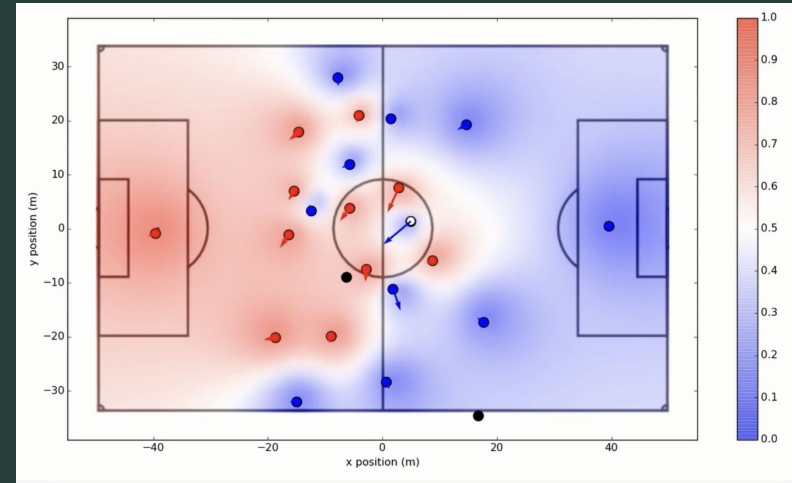
Defines *zones of control* for each frame.

Useful for:

- Evaluating defensive compactness or spacing.
- Understanding support and passing options.
- Measuring pressing intensity (how fast a team closes spaces).

Limitations:

- Instantaneous (single-frame) → no continuous contesting of space.
- Doesn't model how control evolves over time.



https://www.researchgate.net/publication/327139841_Beyond_Expected_Goals

Pitch Control: (Fernández & Bornn, 2018/2019)

“How space control evolves continuously over time”

Builds on Spearman's *time-to-intercept* but models **continuous time** evolution.

Each grid cell has dynamic control probabilities $P_A(t)$, $P_B(t)$:

$$\frac{dP_A}{dt} = \kappa \cdot f_A(t, x, y) \quad , \quad \frac{dP_B}{dt} = \kappa \cdot f_B(t, x, y)$$

where $f_A(t, x, y)$ depends on how soon team A's players can reach that cell.

- Integration continues until both teams can contest the space (≈ 4 s window).
- Output: smooth probabilistic surfaces for both teams (control “accumulation”).

Pitch Control: (Fernández & Bornn, 2018/2019)

Models **how** control emerges rather than just “who gets there first.”

“continuous accumulation of control over time.”

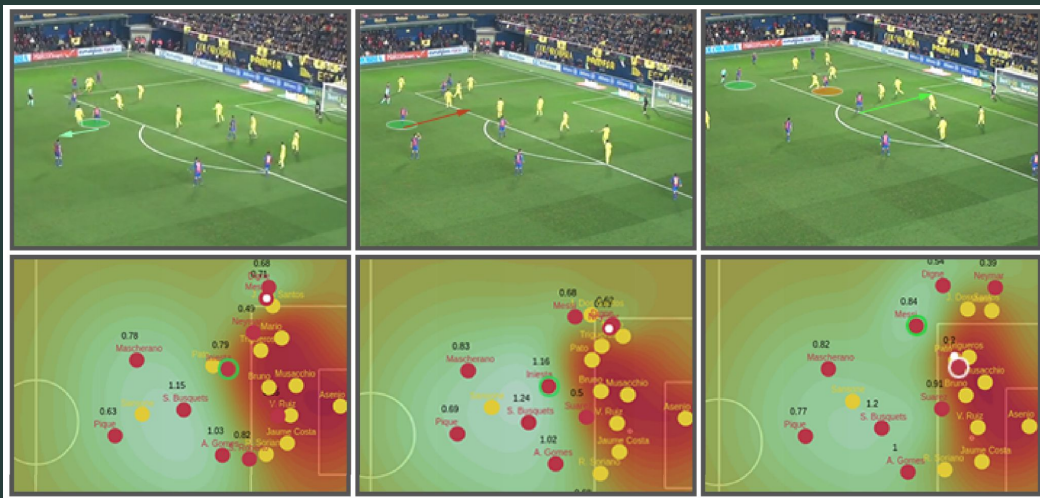
Captures:

- **Shifting control** as players move or press.
- **Transition phases** (e.g., recovering shape after a turnover).

Used in *Expected Possession Value (EPV)* to link space control → outcome probability.

Enables *frame-by-frame valuation*:

$$EPV(x, y) = P_{\text{control}}(x, y) \times E[\text{goal outcome}]$$



https://www.lukebornn.com/papers/fernandez_ssac_2018.pdf

From Instantaneous to Continuous Space Control

- Spearman → *snapshot of reachability*
- Fernández → *dynamic contest over time*
- Foundation for EPV, SoccerMap, and RL-based decision models.

Implementation:

<https://github.com/Friends-of-Tracking-Data-FoTD/LaurieOnTracking/tree/master>

Application: Expected Possession Value (EPV)

Quantify *how much space a team controls* in open play.

Evaluate *pressing, compactness, and defensive balance*.

Combine with xT → **Expected Possession Value (EPV)**:

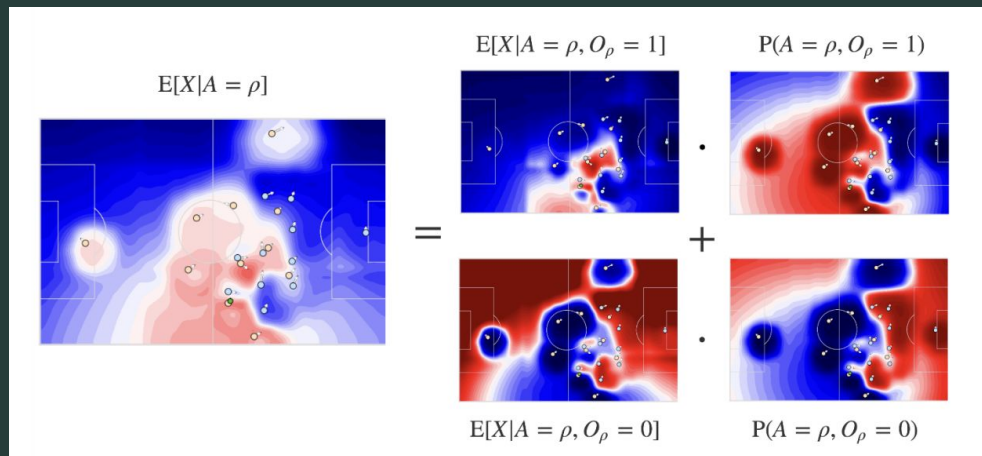
$$EPV = \sum_{x,y} PC(x,y) \cdot xT(x,y)$$

→ measuring how valuable the *controlled area* is, not just ball location.

$$EPV(x,y) = P_{success}(x,y) \cdot V_{success}(x,y) + P_{fail}(x,y) \cdot V_{fail}(x,y)$$

Use case examples:

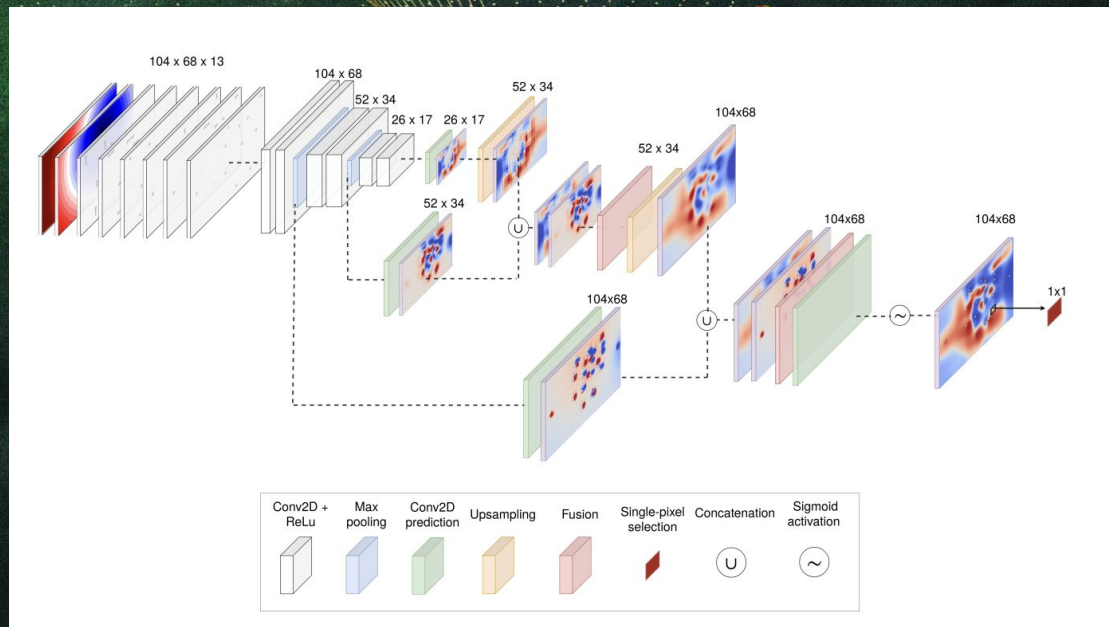
- Identify teams that control high-xT zones off the ball (elite pressing).
- Rank defenders by *threat suppression* via pitch control difference.



Soccermap

SoccerMap: A Deep Learning Architecture for Visually-Interpretable Analysis in Soccer

Javier Fernández^{1,2} and Luke Borna³



SoccerMap (Fernandez & Bornn, 2020)

A fully convolutional neural network architecture that is capable of estimating full probability surfaces of potential passes in soccer, derived from high-frequency spatiotemporal data.

Why it matters:

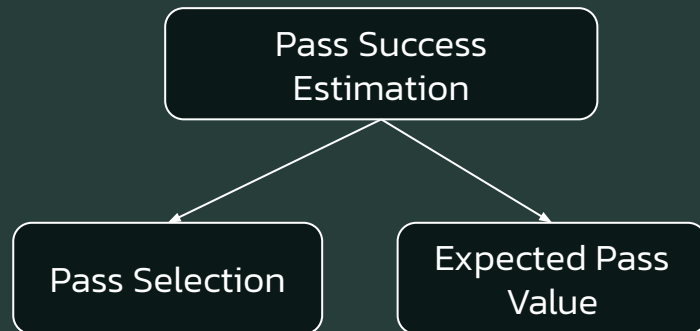
- Learns football-relevant spatial patterns automatically.

Scientific insight:

→ Moves valuation closer to *continuous decision-state modeling*, not discrete events.

Use cases:

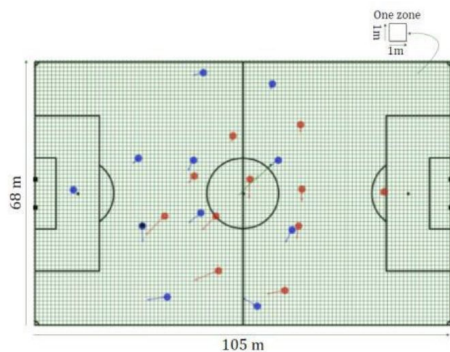
- Quantify *off-ball movement contribution* (e.g., creating space).
- Assess *defensive positioning quality* dynamically.
Compare *team spatial control maps* as tactical fingerprints.



SoccerMap Input Channels

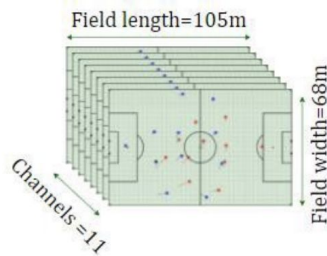
Fine-grained partitioning of the pitch:

Contains $105 \times 68 = 7140$ disjoint zones



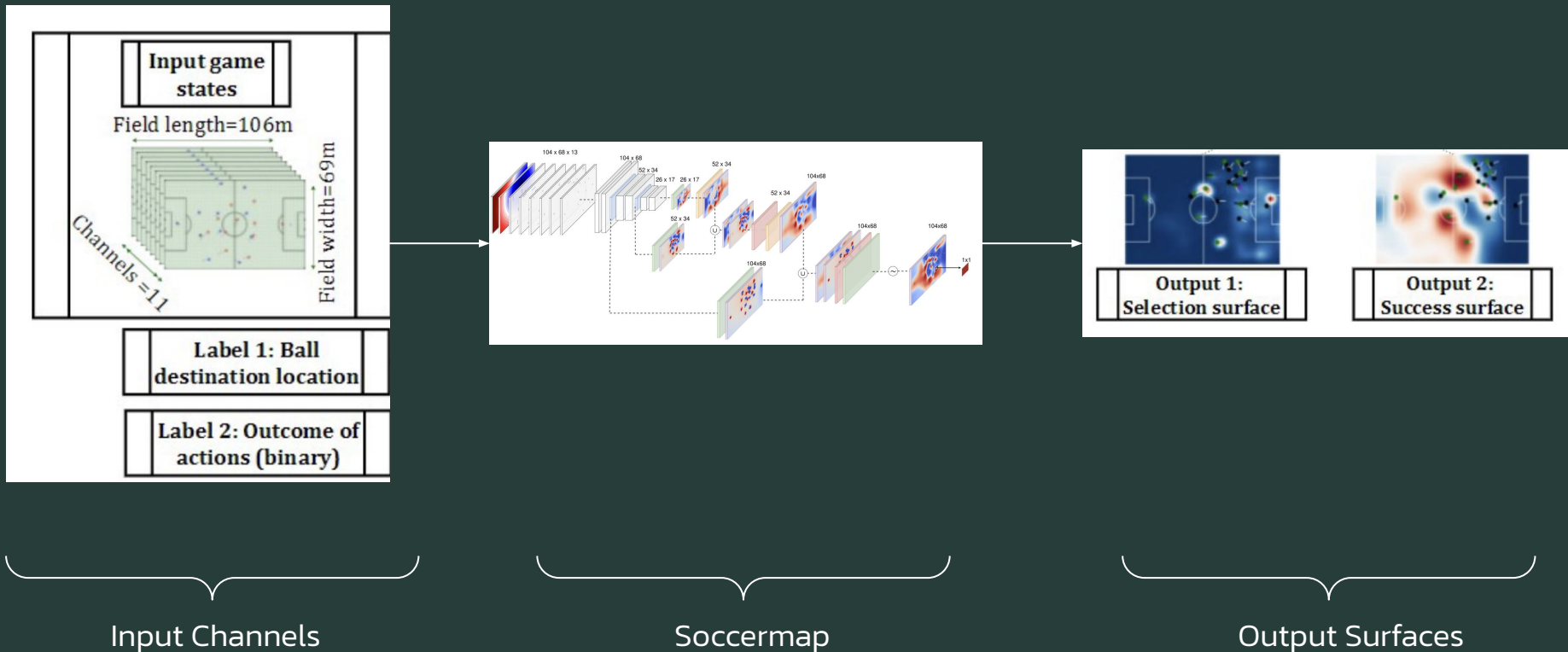
State representation:

(Input shape for a given state to the policy networks)



- Six sparse matrices with the location, and the two components of the velocity vector for the players in both the attacking team and the defending team,.
- Two dense matrices where every location contains the distance to the ball and goal location.
- Two dense matrices containing the sine and cosine of the angle between every location to the goal and the ball location, and one dense matrix containing the angle in radians to the goal location.
- Two sparse matrices containing the sine and cosine of the angle between the velocity vector of the ball carrier and each of the teammates in the attacking team.

SoccerMap Input Channels and Labels



Why SoccerMap is weakly supervised?

In *fully supervised learning*, every input has a **complete target label** for all outputs.

Example:

If your model predicts a **pass probability map** over 12×8 grid cells

$$f(x; \theta) \in [0, 1]^{12 \times 8}$$

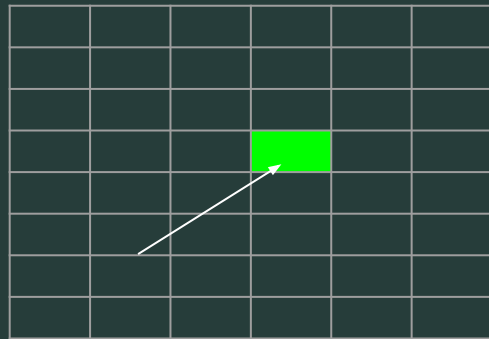
then in *full supervision*, you'd have the **true label** for every *grid cell* — i.e. the true probability that a pass could succeed to each cell.

✓ That means your training data would need to tell you:

“In this situation, passing to cell (3,5) had probability 0.8, to (4,2) had 0.1, ... etc.”

But...

in real football data, **we only observe one pass** — the one that actually happened.



What we actually have

At each time t :

- The input is the **game state tensor** $x_k \in l \times h \times c$ (positions of players, ball, context channels, etc.)
- The only labels we have are:
 - The **destination cell** d_k where the ball was passed.
 - The **binary outcome** $y_k \in \{0, 1\}$ whether the pass succeeded or failed.

So, out of the **entire 12×8 grid of possible destinations**, we only know the label for **one single cell per pass** — the cell d_k

That's why it's **weakly supervised**:

We supervise only *one location* of the full spatial output, not all of them.



How the model is trained

SoccerMap predicts a **full 2D probability map**:

$$f(x_k; \theta) \in [0, 1]^{l \times h}$$

representing the model's belief of pass success probability for *each possible target location*.

But for training, they only take the value at the actual destination cell d_k :

$$f(x_k; \theta)_{d_k}$$

Then they compute the **binary log loss** between that predicted value and the actual outcome:

$$L(f(x_k; \theta), y_k, d_k) = - [y_k \log f(x_k; \theta)_{d_k} + (1 - y_k) \log(1 - f(x_k; \theta)_{d_k})]$$

Intuitive analogy

Imagine showing your model a picture of the whole pitch (the “state”),
and telling it:

“The player passed *here* → (x=55m, y=32m) ... and it was successful.”

That’s the only feedback it gets.

You don’t tell it what would have happened if the player passed anywhere else.

So the model must **learn from many examples** to *implicitly infer* which locations usually succeed given the configuration of players.

Hence:

- ◆ It’s **weakly supervised** (one pixel of supervision per example).
- ◆ It’s **dense prediction** (full map output).
- ◆ It’s **trained with sparse feedback**.

Intuitive analogy

Concept	Fully Supervised	Weakly Supervised (SoccerMap)
Labels per sample	Label for every grid cell	Only one label (the actual pass target)
Model output	Probability for every cell	Probability for every cell
Training loss	Average over all cells	Loss only at destination cell
Intuition	Learn the whole surface directly	Learn the surface indirectly through single samples

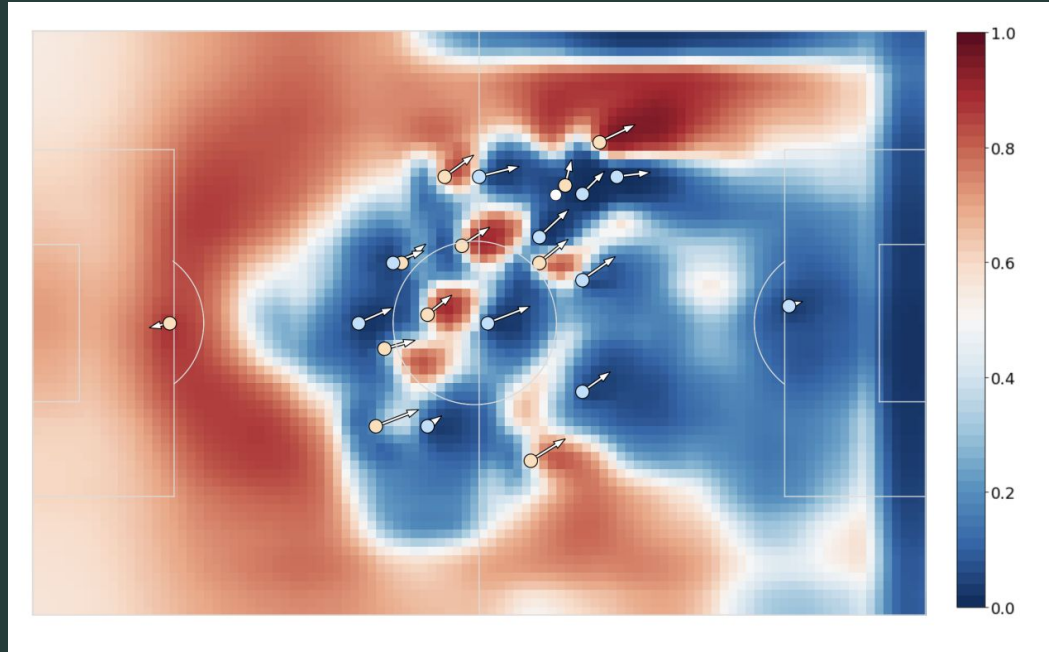
Why it's powerful

- Over thousands of passes, the model gradually **fills in** the probability landscape.
- Even though it's trained only on single cells, it learns that:
 - passes *behind defenders* fail often,
 - passes *into open space* succeed often.

It's like learning to paint the whole picture from dots!!!!



Pass probability surface for a given game situation



Pass Selection & Pass Value Models (SoccerMap Extensions)

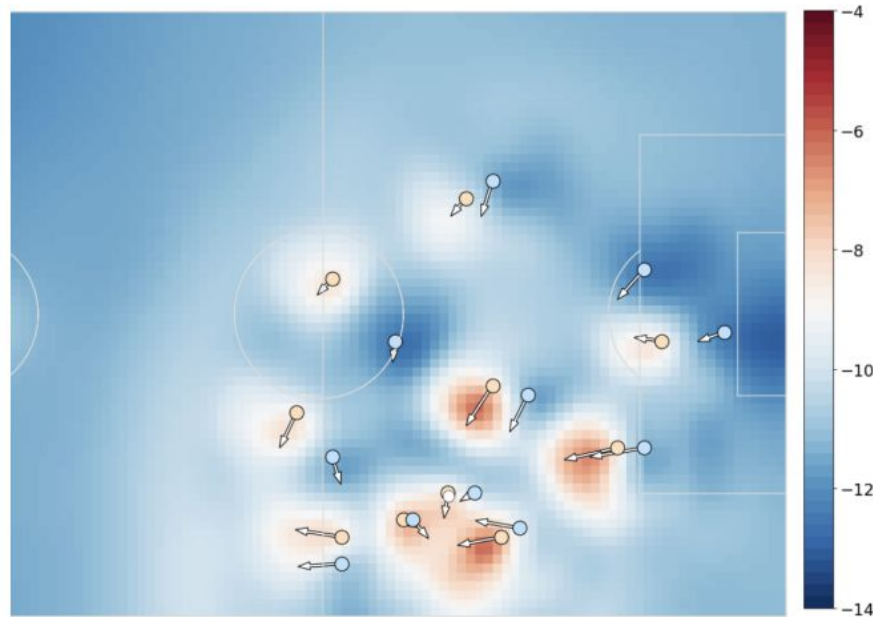
① Pass Selection Model

- **Objective:** Estimate the **likelihood** of a pass being made to **any location** on the pitch.
- **Architecture tweak:**
 - Replace the **sigmoid** activation with a **softmax** layer → ensures all probabilities across the field **sum to 1**.
- **Training target:**
Sparse matrix where **only the actual destination cell** = 1, all others = 0.
- **Output:**
 - A **pass selection probability map** $P_{\text{select}}(x,y)$ showing where a player is most likely to pass next.

② Pass Value Model

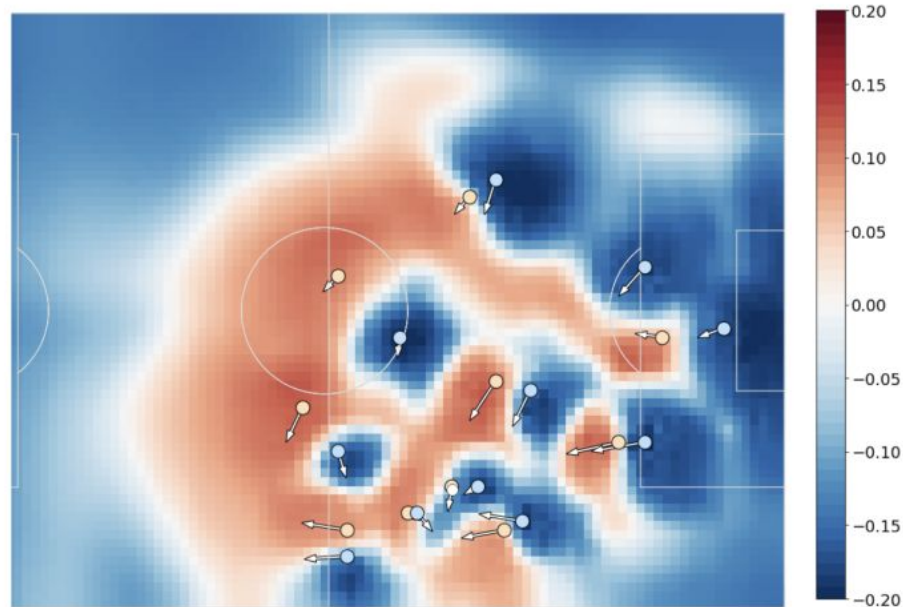
- **Objective:** Estimate the **expected value** (not just success chance) of each possible pass.
- **Motivation:**
 - A low-success pass might still be **highly valuable** if it increases scoring potential.
- **Model adaptation:**
 - Keep SoccerMap structure but **change the target** → use the **expected goals (xG)** value of the **final event in possession**.
 - Update loss function accordingly (continuous regression loss).
- **Output:**
 - A **pass value surface** $V_{\text{pass}}(x,y)$ — how much each potential pass increases expected goals (positive for attack, negative for risk).

Pass Selection & Pass Value Models (SoccerMap Extensions)



Pass Selection

<https://arxiv.org/pdf/2010.10202>



Pass Value

Pass Selection & Pass Value Models (SoccerMap Extensions)

Model Type	Output	Activation	Target	Interpretation
Pass Selection	Probability distribution over all locations	Softmax	Sparse 1-hot map	Likelihood of attempting a pass
Pass Value	Expected value of outcome	Linear / tanh	xG of final event	Strategic quality of the pass

“Players don’t just choose *safe passes* — they choose *valuable ones*. SoccerMap extensions model both **decision likelihood** and **expected reward** of each option.”

Reinforcement Learning for Football Intelligence

Article

<https://doi.org/10.1038/s41467-024-45965-x>

TacticAI: an AI assistant for football tactics

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 Check for updates

Zhe Wang^{1,6}, Petar Veličković^{1,6}, Daniel Hennes^{1,6},
Nenad Tomašević¹, Laurel Prince¹, Michael Kaisers¹, Yoram Bachrach¹,
Romuald Elie¹, Li Kevin Wenliang¹, Federico Piccinini¹, William Spearman²,
Ian Graham³, Jerome Connor¹, Yi Yang⁴, Adrià Recasens¹, Mina Khan¹,
Nathalie Beauguerlange¹, Pablo Sprechmann¹, Pol Moreno¹, Nicolas Hees¹,
Michael Bowling^{1,6}, Demis Hassabis¹ & Karl Tuyls^{1,5}✉

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presented by ESPN 42 ANALYTICS
MARCH 4-5, 2022

Beyond action valuation: A deep reinforcement learning framework for optimizing player decisions in soccer

Pegah Rahimian^{*} · Jan Van Haaren[†] · Togzhan Abzhanova · Laszlo Toka[†]

^{*}Budapest University of Technology and Economics, Hungary – (pegah.rahimian.toka)@tmit.bme.hu

[†]Club Brugge-KU Leuven, Belgium – jan.vanhaaren@kuleuven.be

Data Mining and Knowledge Discovery (2020) 34:1531–1559
<https://doi.org/10.1007/s10618-020-00705-9>



Deep soccer analytics: learning an action-value function for evaluating soccer players

Guiliang Liu¹ · Yudong Luo¹  · Oliver Schulte¹ · Tarak Kharrat²

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APRIL 8-9, 2023 | BOSTON
presented by ESPN 42 ANALYTICS

Leaving Goals on the Pitch: Evaluating Decision Making in Soccer

Maaïke Van Roy, Pieter Robberechts, Wen-Chi Yang, Luc De Raedt, Jesse Davis
KU Leuven, Department of Computer Science; Leuven.AI

Learning to Rate Player Positioning in Soccer

Uwe Dick^{*} and Ulf Brefeld

Abstract

We investigate how to learn functions that rate game situations on a soccer pitch according to their potential to lead to successful attacks. We follow a purely data-driven approach using techniques from deep reinforcement learning to evaluate multiplayer positionings based on positional data. Empirically, the predicted scores highly correlate with dangerousness of actual situations and show that rating of player positioning without expert knowledge is possible.

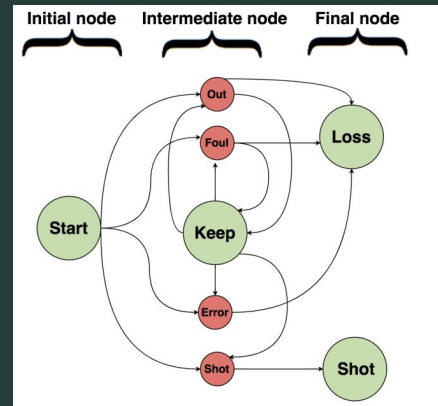
Keywords: deep learning; spatiotemporal data; reinforcement learning; scoring function

Football as a Markov Decision Process (MDP)

Core Components of a Markov Decision Process (MDP)

An **MDP** is defined by five key elements:

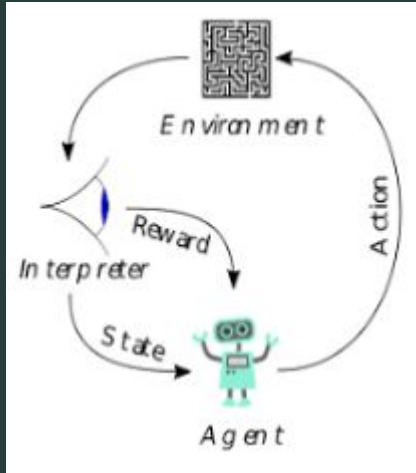
- **State space (S)**: Represents all possible situations or configurations the environment can be in.
- **Action space (A)**: Defines the set of actions or decisions available to the agent in each state.
- **Transition function (P)**: Describes the probability of moving from one state to another after taking a specific action.
- **Policy (π)**: Specifies the decision rule — the probability of selecting each action given the current state.
- **Reward function (R)**: Assigns an immediate numerical reward (or penalty) for each state–action–next state transition, guiding the learning process.



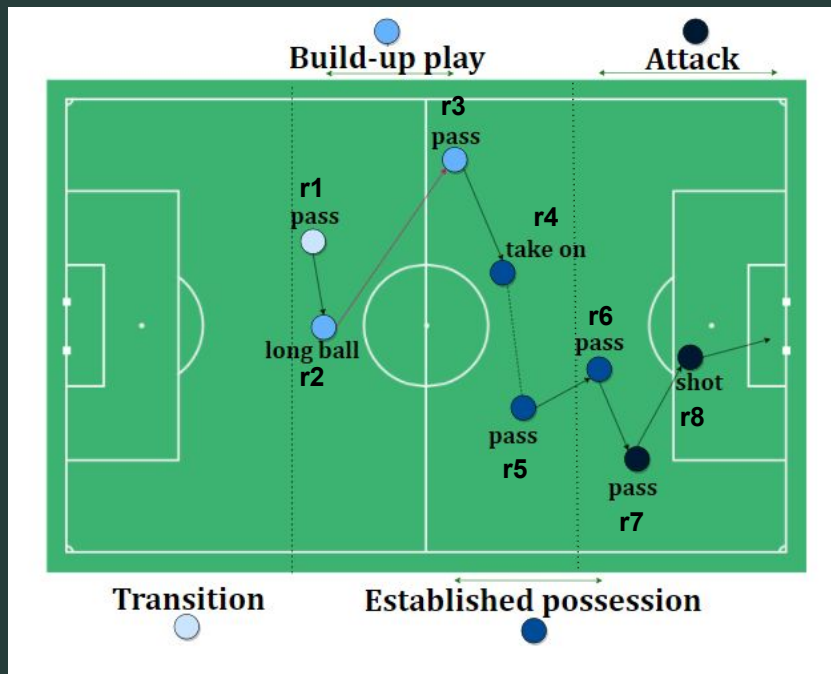
Reinforcement Learning Goal

Learn an **optimal policy** π^* that maximizes expected cumulative reward:

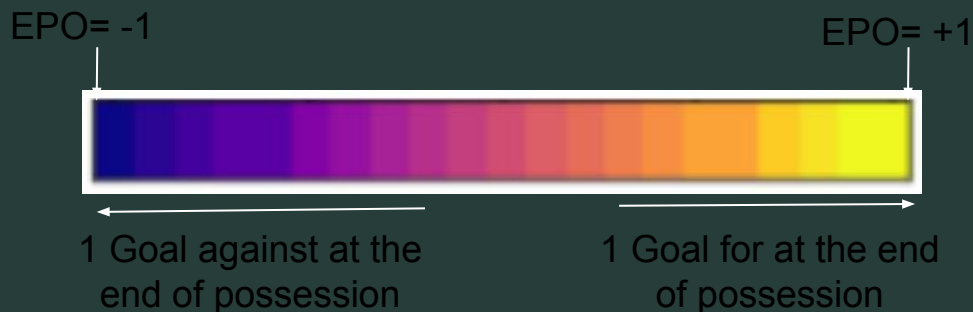
$$\pi^* = \arg \max_{\pi} \mathbb{E} \left[\sum_t \gamma^t r_t \right]$$



Expected Possession Outcome (EPO)

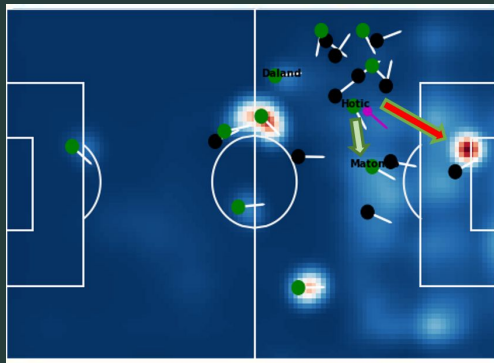


$$EPO = \sum_{t=1}^H \mathbb{E}_{s_t, a_t \sim \pi} [\gamma^t r(s_t, a_t)]$$

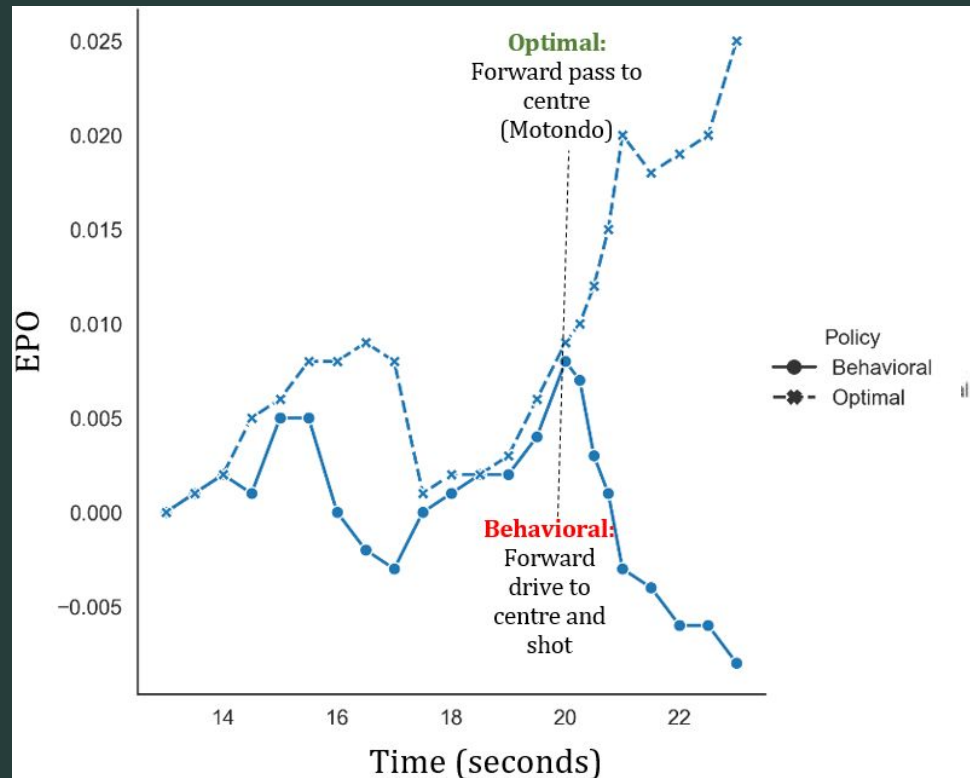
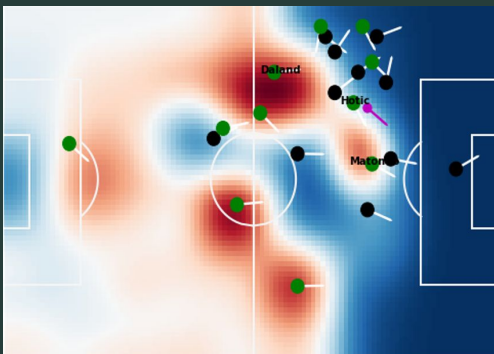


Which action could maximize the scoring chance?

Selection Surface

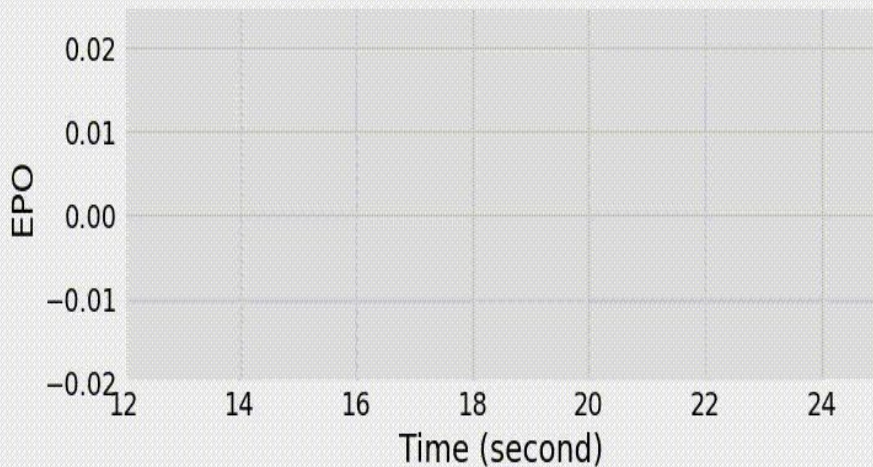


Success Surface (Reward)



The effect of a bad decision on losing the goal scoring chance

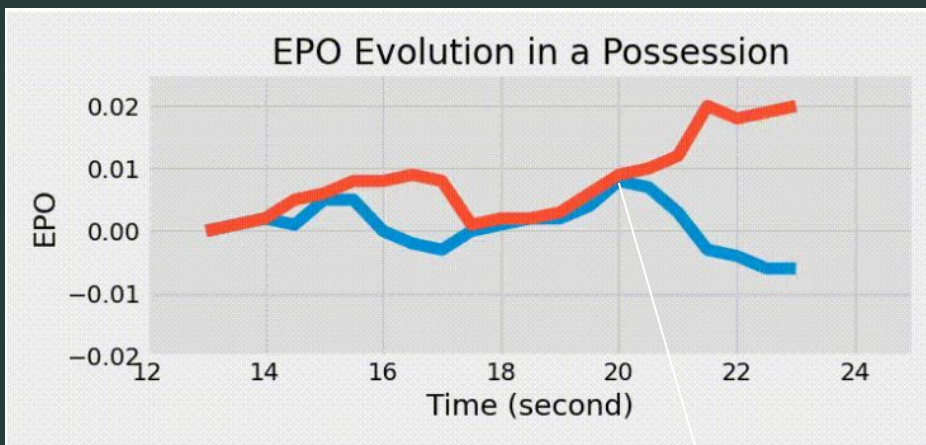
EPO Evolution in a Possession



unsuccess_pass



The effect of a bad decision on losing the goal scoring chance



Bad decision!

Markov Chain xT vs. Reinforcement Learning

Aspect	Markov Chain xT	Reinforcement Learning
Transition Model	Fixed, estimated from data	Learned via reward optimization
Action Evaluation	Expected threat from state $\rightarrow$ next state	Cumulative expected return over future actions
Goal	Estimate value of current location	Learn optimal policy to improve decisions
Learning Type	Passive (no control)	Active (optimize policy)
Analogy	"How valuable is this position?"	"Which action should I take here?"

 In short:

- xT tells *how threatening a location is*.
- RL tells *what the player should do next* to maximize expected reward.

Offline Reinforcement Learning for Football

⚙️ Why Offline?

- Football data is *pre-recorded* (no online exploration possible).
- RL must **learn from logged trajectories** of human players (event and tracking data).

■ Offline RL Objective:

Learn a **policy** $\pi(a|s)$ that performs better than the behavior policy (real player actions),

using only historical data:

$$\max_{\pi} \mathbb{E}_{(s,a,r,s') \sim D} \left[Q_{\pi}(s, a) \right]$$

Offline RL Methods Used in Football

Method	Concept	Use Case
Behavior Cloning (BC)	Supervised imitation of player decisions	Learn realistic baseline policy
Batch RL *	Avoid extrapolation errors	Safe policy improvement
Conservative Q-Learning **	Penalize overestimation of unseen actions	Evaluate unseen strategies
TD3+BC ***	Combine actor-critic learning with imitation	Optimize policy under real data constraints

d3rlpy is one of the best offline RL libraries: <https://d3rlpy.readthedocs.io/en/v2.8.1/>

* <https://pubmed.ncbi.nlm.nih.gov/30672712/>

** <https://link.springer.com/article/10.1007/s10994-024-06611-1>

*** <https://link.springer.com/article/10.1007/s10618-020-00705-9>

RL for Action Valuation and Policy Optimization

Applications:

-  *Action Valuation*: learn $Q(s,a)$ — expected long-term value of each decision (pass, dribble, shot).
-  *Policy Optimization*: simulate “what-if” scenarios (e.g., would another pass increase success rate?).
-  *Training AI coaches*: model tactical decisions that maximize expected goal difference.

Football Example:

In a given state (player under pressure, ball on right flank):

- Markov xT → tells “crossing has xT = 0.03.”
- RL → tells “crossing now maximizes long-term xG expectation by +0.015 vs. short pass.”

Practical Football Use Cases

Use Case	Example
Tactical optimization	Simulate best passing strategies under pressing
Player decision modeling	Quantify risk/reward tendencies
Team behavior learning	Learn policies that reflect club philosophy
Counterfactual scouting	Estimate how a player would perform in another team's system
Game simulation	Generate AI vs. human tactical scenarios

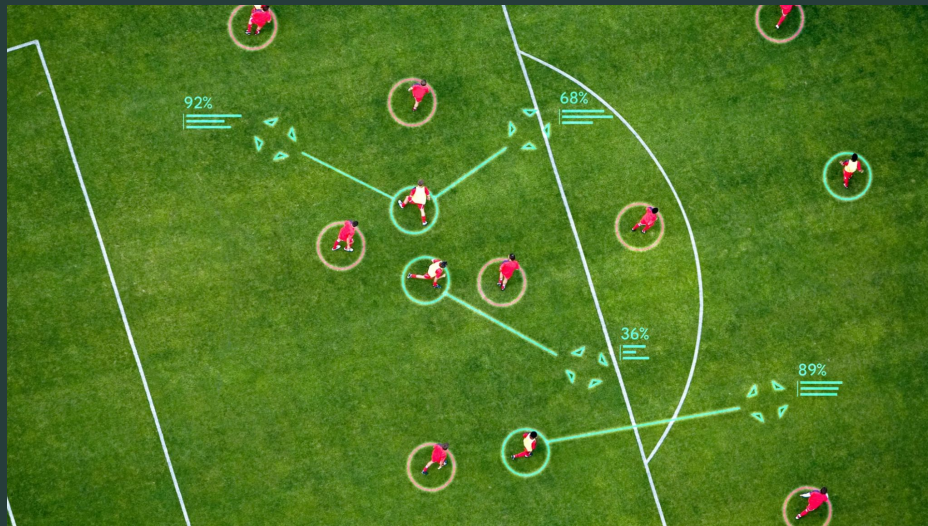
TacticAI — Reinforcement Learning for Tactical Intelligence in Football

Developed by: Google DeepMind × Liverpool FC (2024)

Objective: Optimize *set-piece strategies* (e.g., corners) through data-driven simulation and generative AI.

1 What is TacticAI?

- A **neural network–based tactical recommender system** built on **reinforcement learning** principles.
- Models *player movement, positioning, and passing options* during **corner kicks**.
- Trained on **event + tracking data** from real matches.



<https://deepmind.google/discover/blog/tacticalai-ai-assistant-for-football-tactics/>

TacticAI — How it works?

Component	Description
Input	Player trajectories, ball position, team formation before the corner
Architecture	Graph Neural Network (GNN) capturing spatial-temporal interactions
Learning Objective	Predicts the probability of each outcome (e.g., shot, goal, clearance)
Optimization	Suggests alternative player positions maximizing expected success (offline RL style)

- Moves from *retrospective* analysis (e.g., VAEP, xT) → to *prescriptive* optimization.
- Helps coaches **simulate “what-if” tactical setups** under different positioning.
- Shows how **offline reinforcement learning + generative modeling** can assist human decision-making.

The background is a dark green, crumpled paper texture. Scattered across the top and right sides are numerous small, golden, crescent-shaped confetti pieces. At the bottom of the image, there are several thin, golden, wavy lines that sweep across the frame, accompanied by a pattern of small golden dots.

Thank You!